

Research paper

Multidisciplinary design and optimization of winged architectures for reusable launch vehicles

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ABSTRACT

Partially reusable launch vehicle design has raised a great interest in order to minimize the costs of space transportation by recovering and refurbishing the first stage. Several configurations such as toss-back architectures are now operational. These concepts require additional rocket propellant in order to carry the boost-back and landing maneuvers, that induce several losses in terms of payload mass performance for the ascent mission. In order to limit these losses, this paper focuses on the design of two types of architectures to carry out the return-to-launch-site mission. The first, named glide-back, uses the main propulsion system of the first stage to perform a boost-back burn and returns to the landing site by a gliding mode using additional lifting surfaces. The second, named, fly-back, performs the return-to-launch-site mission using several air-breathing engines located in the nose of the first stage combined with lifting surfaces. In this paper, the design of a reusability kit, allowing to provide the first stage with both expendable and reusable capabilities, is investigated. This kit is composed of the lifting surfaces, the nose including the air-breathing propulsive system (for fly-back configuration), landing gears and additional avionics. This paper addresses the design of such reusability kits and presents the mission specifications, the design process relying on Multidisciplinary Design Optimization techniques, the vehicle performance and the optimal trajectories. Different analyses (aero-propulsive, optimal control, operational considerations, etc.) are detailed and trade-offs between the studied configurations are assessed.

1. Introduction

Reusable launch vehicle design has raised a great interest in the last decade in order to minimize the costs of space transportation by recovering and refurbishing the first stage [1]. Toss-back configurations are operational and their efficiency is now well established [1,2]. This type of concepts consists for the first stage in performing a Return To Launch Site (RTL) mission by using the main rocket engines propulsion and landing vertically. To do so, this architecture has to carry out additional rocket propellant in the ascent phase in order to perform the boost-back and landing burns [3]. As a consequence, this induces several losses in terms of performance for the ascent mission and reduces the payload capability at the injection. In order to limit these losses, different alternative configurations can be investigated [4]. For example, it may be interesting to reduce the additional rocket propellant that has to be

used for the RTL mission. It can be partially performed using aeronautical technologies (lifting surfaces, air-breathing propulsion, etc.) [5–7]. This paper focuses on the design of winged configurations that allow to limit or avoid the need of additional rocket propellant to perform the RTL mission. The considered architectures lift-off vertically and land horizontally similarly to an aircraft. The first configuration, named glide-back [8], combines both space and aeronautical technologies to recover the vehicle first stage. After the first stage ascent mission, the second stage is jettisoned and the first stage reignites a part of its rocket engines to cancel its horizontal velocity. Then, the vehicle performs an aerodynamical reentry and glides back to the landing site. For that purpose, aerodynamical nose, lifting surfaces, gears and corresponding power avionics are added to the initial stage configuration.

The second architecture, named fly-back [9], uses only aeronautical technologies for the RTL mission and allows to completely avoid

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rocket propellant for this phase. After its separation with the second stage, the first stage carries out a ballistic phase followed by a high angle of attack atmospheric reentry. Then, several turbojets, that are located in the nose of the first stage, are ignited to carry out a cruise flight and to land the vehicle horizontally. In this case, in addition with the previous aeronautical elements, several turbojets and their propulsive systems are included. Different glide-back and fly-back configurations have been proposed and studied in the literature such as: Future Launcher System [10], Everest [11], Liquid Fly-Back Booster [12], Barguzin [13], Spaceliner [14], etc. For more details on the recent studied European configurations of reusable launch vehicle, please refer to [4].

The novelty in this paper in terms of launch vehicle architectures is the design of reusability kits, allowing to provide the first stage with both expendable and reusable capabilities. These kits, composed of the lifting surfaces, the nose (including the airbreathing propulsive system for fly-back configuration) and additional subsystems (e.g., landing gears), can be mounted on the main core of the launch vehicle for performing several reusable missions, and then removed and installed on another first stage if the current one is used for a last expendable mission. In that way, the reusability kits may be used a large number of times allowing to lower the costs. The design is performed using multidisciplinary design optimization techniques [15,16] allowing to assess the overall launch vehicle performance and to meet optimal compromises between the different disciplines (propulsion, aerodynamics, structure, trajectory, etc.) as well as the different flight phases (ascent, re-entry and return to launch site). Trade-offs between the two types of configurations are assessed regarding the performance of the RTLS phase and the penalty induced by the reusability kit for the ascent mission. This study is a collaboration between the French Space Agency (Centre National d'Études Spatiales - CNES) and the French Aerospace Lab (Office National d'Études et de Recherches Aérospatiales - ONERA) on Reusable Launch Vehicles (RLV).

The rest of the paper is organized as follows: in Section 2, the mission specifications and the technological choices regarding the considered launch vehicle are described. Section 3 presents the multidisciplinary process and the disciplinary models used to assess the launch vehicle performances and to allow architecture comparisons. The trade-off assessment between glide-back and fly-back configurations is carried out in Section 4 by presenting the obtained optimal architectures and associated optimal trajectories. In addition, in Section 5, the fly-back configuration is further investigated with different disciplinary model improvements as this configuration presents the highest potential in comparison with the glide-back.

2. Description of mission specifications, technological choices and reusability kits

This study deals with the design of a Two-Stage-To-Orbit (TSTO) launch vehicle with a reusable first stage. For that purpose, an expendable configuration is modified thanks to a reusability kit in order to provide it with reusability capability. For the main core, the Ariane-Next configuration [17] has been chosen as a baseline. The diameter of the stages is considered as fixed and is equal to 4.6 m. The first stage is powered by seven PROMETHEUS (Precursor Reusable Oxygen METHane cost Effective propUlsion System) engines [18] using liquid oxygen and liquid methane (LOx/LCH₄) propellants. The second stage is also powered by one PROMETHEUS engine. The reference mission consists in injecting a 6 tons payload into a 800 km circular Sun Synchronous Orbit (SSO). The launch site is the European Space Port in French Guiana. Additional constraints are considered in order to take operational conditions into account. Different thresholds concerning the loads (axial and transverse load factors, maximum dynamic pressure, maximal aerothermal flux) during both the ascent and return phases are considered (Table 1). Moreover, visibility and safety specifications (allowed flyover zones) near the launch and landing sites are taken into account (Fig. 1).



Fig. 1. Allowed flyover zone for ascent phase and landing, with in red the location of the landing runway. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

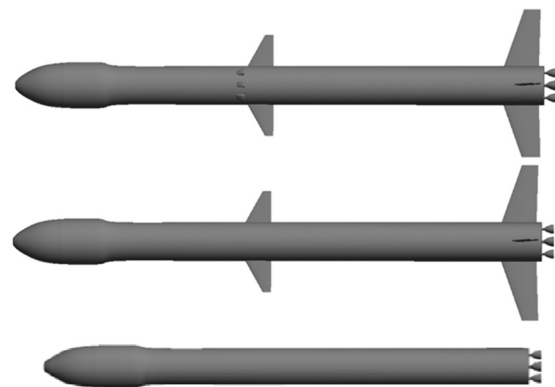


Fig. 2. Expendable configuration (bottom), glide-back configuration (center) and fly-back configuration (top).



Fig. 3. Glide-back architecture first stage with in black the reusability kit.

To design the reusable architectures, an expendable configuration is modified thanks to a reusability kit in order to provide it with reusability capability. This reusability kit (Fig. 3 for the glide-back configuration) is composed of lifting surfaces such as main wings, canards and a vertical plane, front and rear landing gears. The wings and rear landing gears are located in a case that is attached to the thrust frame and the inter-tank of the main core. The canards, aerodynamic nose and front gears are located in a reusability pack (composed of the nose and an additional skirt) that is positioned in the front of the first stage. This reusability kit can be removed for expendable missions. For the fly-back architecture, additional turbojets and corresponding propulsive systems are included in the reusability kit and are located in the nose of the first stage. The considered vehicles are Vertical Take-Off - Horizontal Landing (VTHL) architectures. Fig. 2 illustrates the glide-back and fly-back configurations of the first stage, with respect to the classical expendable one.

Numerous trade-offs between technological choices have to be assessed in order to find the optimal configurations. Among them, the

Table 1
Characteristics of the mission and specifications.

Characteristics of the mission		
Orbit	Sun Synchronous Orbit at 800 km of altitude	
Payload mass	6 tons	
Architecture	Two-Stage-To-Orbit	
Launch and landing sites	European Space Port, Kourou, French Guiana	
Specifications	Ascent phase	RTLS phase
Max dynamic pressure (kPa)	40	110
Max axial load factor (-)	4.5	8
Max transverse load factor (-)	0.3	6
Max aerothermal flux (kW/m ²)	100	140
Visibility (-)	Visibility from telemetry stations	
Safety (-)	Flyover zones above the European Space Center to respect	

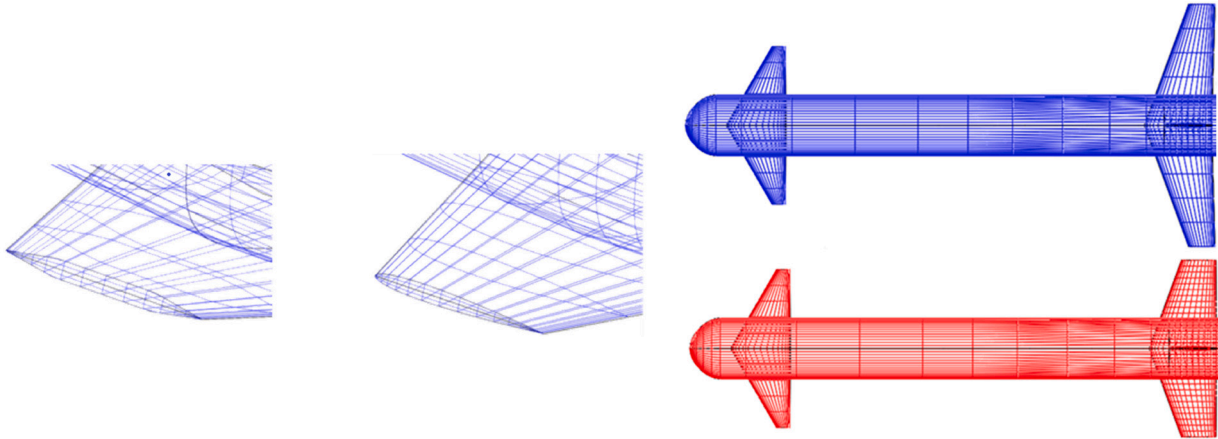


Fig. 4. Different technological choices for wing planforms and airfoil profiles (double-wedge — left and NACA — right).

main technological choices deal with the lifting surfaces design (*i.e.*, wings, vertical and canards planforms, aerodynamic profiles, location of the lifting surfaces - Fig. 4), number, location and types of turbojets for the fly-back configuration. For the glide-back configuration, the apportion of rocket propellant that will be used for the boost-back burn is also a key driver. For the fly-back architecture, an analog design parameter is the amount of kerosene for the RTLS cruise flight. All the technological choices will be described in details in Section 4.

The ascent phase of the two reusable architectures is the same as for an expendable launch vehicle. After a vertical lift-off, the launch vehicle performs a pitch-over maneuver followed by a gravity turn maneuver in order to limit the aerodynamics losses. Then, once the dynamic pressure is sufficiently low, an optimized control-law is involved in order to reach the transfer orbit. Once the altitude and velocity conditions of the transfer orbit are reached, the engine of the second stage is cut-off for the coasting phase. This engine is reignited a last time for a circularization burn allowing to inject the payload on the target orbit.

Concerning the RTLS phase of the glide-back architecture (left of Fig. 5), once the second stage is jettisoned, the first stage performs a RTLS trajectory composed of the following steps :

- turn around phase: this phase allows to place the stage in an appropriate orientation using for instance cold gas thrusters for the boost-back burn;
- boost-back maneuver: in this phase, the main rocket engines are re-ignited to invert the velocity vector and to provide an impulse in the direction of the landing site (only 3 PROMETHEUS engines are used for this maneuver);
- atmospheric re-entry: this phase follows the boost-back maneuver and performs a controlled re-entry to limit the different loads (thermal, dynamic pressure, etc.);

- pull-up maneuver and glide phases: after re-entry, the first stage performs a pull-up maneuver and a glide phase to exploit the lifting surfaces to reach the landing site, maximizing the lift-over-drag ratio;
- final phase: this phase is composed of a turn to align with the runway while satisfying the visibility and safety constraints.

The flight phases of the fly-back architecture for the RTLS mission (right of Fig. 5) are described in the following:

- ballistic flight: the first stage performs a full ballistic flight with an angle of attack of zero degree in the upper layers of the atmosphere;
- atmospheric re-entry: this phase follows the ballistic flight phase and performs a re-entry with a high angle of attack in order to limit the loads and temperature;
- U-turn and pull-up maneuvers: after the re-entry, the first stage performs a U-turn maneuver in order to go back to the landing site and a pull up maneuver to reach the conditions of turbojet ignition;
- final phase: this phase is a powered phase using the air-breathing engines (turbojets) located in the nose of the vehicle. This phase is analog to an aircraft cruise phase at subsonic Mach number.

In order to assess the performances of the two types of configurations, a multidisciplinary process is implemented involving different coupled disciplinary models. The multidisciplinary process is described in the following section.

3. Description of multidisciplinary process

3.1. Disciplinary models

In this section, all the disciplines that have been included in the multidisciplinary design process are presented.

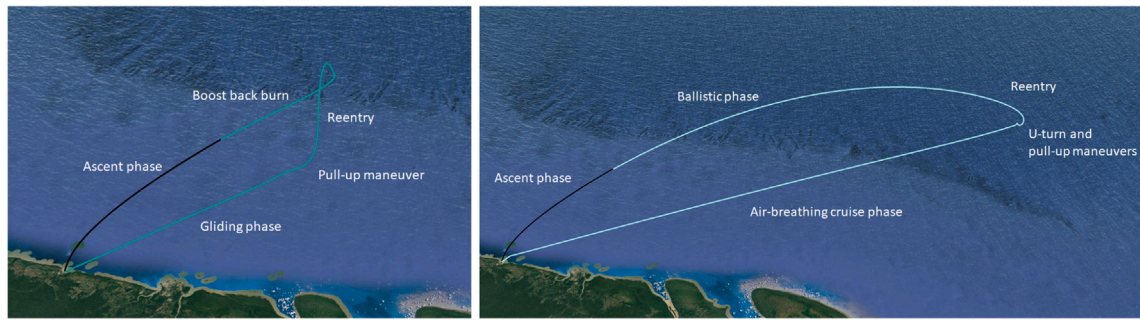


Fig. 5. Flight phases of glide-back architecture (left) and fly-back architecture (right).

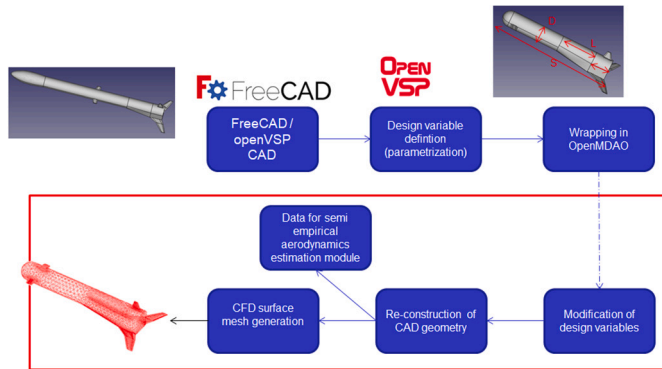


Fig. 6. Illustration of geometry discipline.

3.1.1. Geometry and structural design

The geometry and structural design discipline is responsible for defining the overall geometry of all the stages considering different key geometrical parameters. Moreover, it is in charge to generate the vehicle surface meshes that will be used for the aerodynamics assessment. Eventually, the discipline computes the mass budget of the launch vehicle. This discipline uses a parametric geometric modeler to define the global vehicle geometry from several decision variables such as the length of the tanks, the wing planform, the diameter of the stage, *etc.* This geometry is used to build a preliminary Computer-Aided Design (CAD) that is integrated in the multidisciplinary process and returns as outputs the vehicle surface meshes for aerodynamic performance assessment (Fig. 6). The geometry model relies on FreeCAD [19] and OpenVSP [20] to carry out the parametric definition of the geometry and to provide CAD tools for the mesh generations.

From the geometry data and the loads given by the trajectory, a structural design of the first stage is performed using a dedicated module developed at ONERA. In general, it is necessary to distinguish between the primary structures (wings, tanks, skirts, thrust frame, *etc.*) and secondary structures (engines, equipments, *etc.*). Primary structures aim at taking up all the loads (thrust, aerodynamics, inertial) whereas the secondary structures are not a part of the structural backbone of the launcher.

The primary structures are designed using both analytical formula and Finite Element Analysis (Fig. 7) to consider key elements such as the buckling, bending moment, wing deflection, *etc.* When the geometry of a component is complex (for example the thrust frame), analytical sizing methods can be only used at a local level. The global sizing of the component has to be performed with finite element methods. Consequently, the sizing module couples a large variety of sizing methods depending on the component to evaluate. Especially, the thrust frame has been modeled using finite elements with beam components. The engine attachment is also modeled using a beam component. The sizing process of this component consists in redesigning all the

elements of the thrust frame until a convergence criterion regarding the mass is reached. The tanks and skirts are mainly designed with respect to a buckling criterion as described in [21]. They are based on aluminum alloy to enhance their mechanical properties with respect to classical alloys. In particular, the tanks need to be pressurized for the RTLS phase to be compliant with buckling. Furthermore, the sizing of lifting surfaces (*e.g.*, canards, wings) is performed using finite element analyses to take into account the aerodynamic loads at different flight points. Considering the relatively small relative thickness of the lifting surfaces, specific internal structure is considered in order to increase the buckling resistance without penalizing the overall mass of the component.

All the loads for the different phases (ascent and return) are considered (coming from the aerodynamics discipline or the propulsion for instance). The considered load cases to design the primary structures depend on the vehicle parts and the different phases during the flight (*e.g.*, lift-off, atmospheric reentry, pull-over maneuver). The most critical load case is considered for each vehicle primary structure allowing to have a consistent structural design of the launch vehicle with the encountered flight loads. Concerning the sizing process of the primary structural elements, internal design variables are optimized (*e.g.*, for the tanks: the thickness, the stiffener dimension and separation distance, for the wings: the spars and the ribs geometries and thicknesses) in order to minimize the structural mass while satisfying specific constraints such as global and local bucklings or maximum allowed deformation. Specific penalty masses are included for each structural component to account for attachment mechanisms, weldings, *etc.* based on legacy guidelines from the literature [21].

The secondary structures (landing gears, avionics, equipments, turbojets, *etc.*) are either provided by the ArianeNext configuration, either taken from manufacturers data for the turbojets, or estimated using Mass Estimation Relationships (MER) formula [22]. The second stage is a classical expendable one. Consequently, standard MERs have been used to determine its mass. This module is included in the multidisciplinary design optimization process using OpenMDAO [23] framework as a parametric function allowing to reestimate the mass budget for the different values of the design variables controlled by the optimization process.

3.1.2. Aerodynamics

In order to be able to carry out global performance assessment, the aerodynamic coefficients of the vehicles have to be accurately computed for different architectures (entire launch vehicle, first stage only with glide-back reusability kit, first stage only with fly-back reusability kit, second stage only) and different flight phases (ascent phase: subsonic, supersonic, hypersonic, reentry phase: high angle of attack flight, return phase: subsonic, supersonic, powered phases or not). This makes the design of the aerodynamic module challenging and dedicated tools of different fidelities have been combined. Furthermore, this module has to be evaluated a large number of times in the optimization process for different aerodynamic configurations. In that way, the computational cost is another key driver for the aerodynamic assessment.

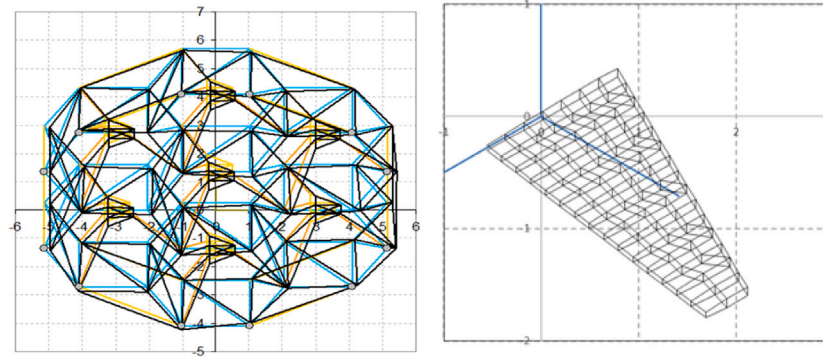


Fig. 7. Thrust frame and wing design.

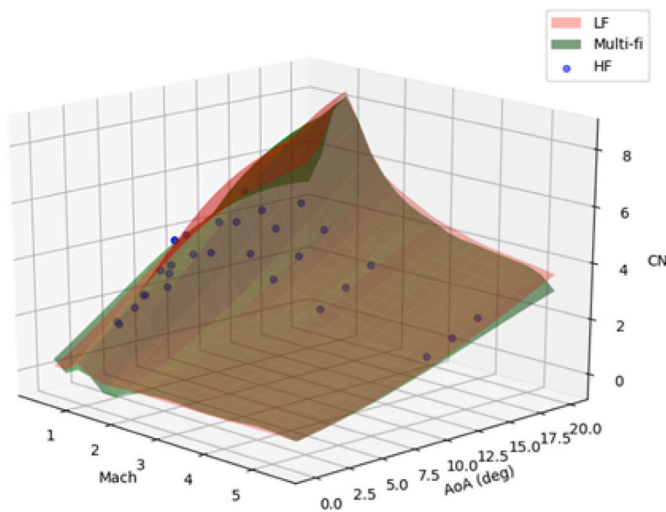


Fig. 8. Multi-fidelity techniques for aerodynamics for a reusable launch vehicle configuration (Low fidelity model in red, High fidelity data in dot and multi-fidelity model in green). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

To estimate the aerodynamic characteristics of the studied configurations with an affordable computational cost, multi-fidelity techniques are used [24–26]. The multi-fidelity model relies on Gaussian process (Fig. 8). These multi-fidelity techniques allow to aggregate the responses of the different aerodynamic codes depending on the flight phase and the flight conditions (Mach number, angle of attack) while accounting for their accuracies and uncertainty levels.

Four levels of fidelity have been used in the aerodynamics discipline. The first one is a semi-empirical code named MISSILE [27] that allows to compute the aerodynamics characteristics from Mach 0. to Mach 10. This code involves analytical formula calibrated with wind tunnels results. The second code is an in-house code named SHAMAN which is based on Local Surface Inclination (LSI) methods [28–30] and allows to determine the aerodynamics coefficients for supersonic and hypersonic regimes at a reduced computational cost (Fig. 9).

To improve the accuracy of the aerodynamic coefficients estimation, Euler CFD calculations are also used thanks to an aerodynamic module based on the ONERA CANOE [31] module relying on SU2 [32]. Eventually, a RANS CFD model based on elsA [33] is used for a calculation campaign (Fig. 10) to improve the accuracy of the coefficients estimates for specific flight domains such as the reentry phase. These models have been validated on classical launch vehicle configurations.

3.1.3. Propulsion

Concerning the rocket propulsion, the two stages are powered by PROMETHEUS engines using LOx/LCH₄ propellants [34] currently

under development in Europe. The engine cycle is a gas generator cycle composed of a mono-shaft turbopump, a gas generator, two chamber valves, two gas generator valves and a regeneratively cooled combustion chamber [18]. The first stage involves seven engines and the second stage uses one engine. The data of the engine have been provided by CNES for the study. These engines are throttleable and re-ignitable. The thrust varies from 300 kN to 1000 kN. The control of the engine with respect to the thrust domain is based on a full electrical valves actuation. The throttling level is optimized during the trajectory in order to optimize the performance and to meet the design specifications (e.g., maximal dynamic pressure).

Concerning the air-breathing propulsion, semi-empirical formula taken from [35] are implemented in order to determine the main performances (the thrust and specific fuel consumption) as functions of the Mach number and the altitude (Fig. 11). Two military turbojet engines M88 and EJ200 are modeled from their main characteristics (nominal thrust, pressures ratio, by-pass ratio, etc.). This model has been calibrated on the manufacturer data.

3.1.4. Trajectory

The trajectory discipline is responsible for the assessment of the overall performance of the launch vehicle. To do so, all the flight phases are simulated. The trajectory is optimized globally (considering all the phases together) using a direct control strategy [36], i.e., the control laws are parameterized by a set of control variables which are optimized as well as the other discipline design variables. This strategy has proven its efficiency in launch vehicle Multidisciplinary Design Analysis and Optimization (MDAO) because of its large robustness with respect to the other discipline parameters [15]. The three degrees-of-freedom equations of motion are integrated with a 5th order variable step Runge–Kutta integrator. Different events such as the beginning of the exo-atmospheric flight, the jettisoning of the fairing, the coasting phase, etc. are considered in the trajectory integration. The parametrization of the different phases of the trajectory is presented in the following paragraphs.

Ascent trajectory. The objective of the ascent phase optimization is to meet the orbit conditions while minimizing the propellant consumption. Several constraints about the maximal values of the dynamic pressure, the angle of attack and the axial load factor, but also the visibility and safety have to be taken into account. The ascent phase is carried out at a very low angle of attack to minimize the influence of the lifting surfaces during that phase. The ascent trajectory is the same for the two studied architectures decomposed into a vertical flight, following by a pitch over maneuver and a gravity turn phase during the atmospheric flight. When the exo-atmospheric conditions are met, a controlled phase is involved to reach the orbit conditions. To do so, a parameterized bi-linear tangent law for the pitch angle is optimized [22]. All the parameters of the different phases are optimized during the design process. The second stage performs a coast-phase

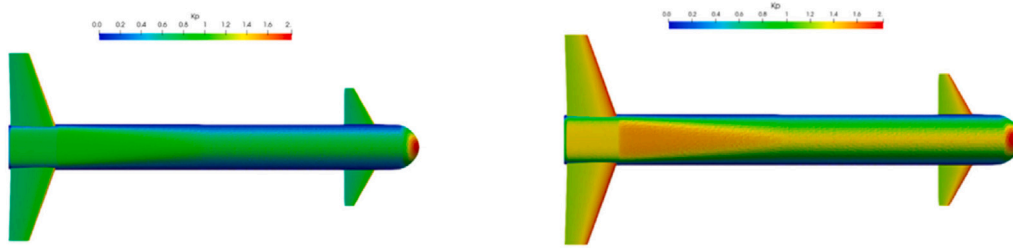


Fig. 9. Results from aerodynamic assessment using LSI methods.

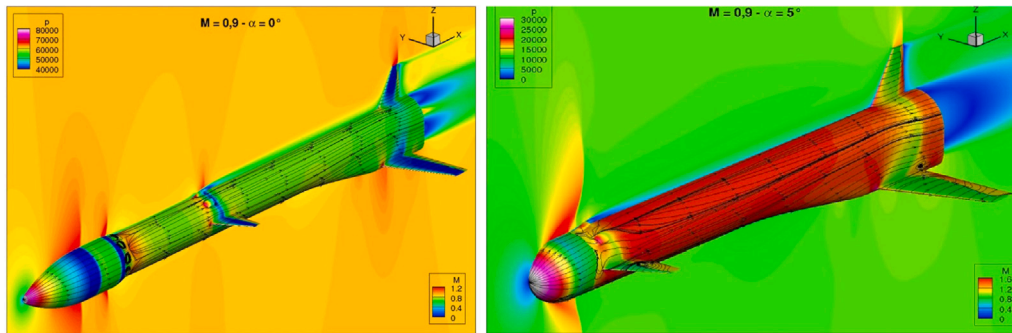


Fig. 10. Examples of CFD RANS calculations (left: global vehicle in ascent phase, right: first stage in RTLS phase).

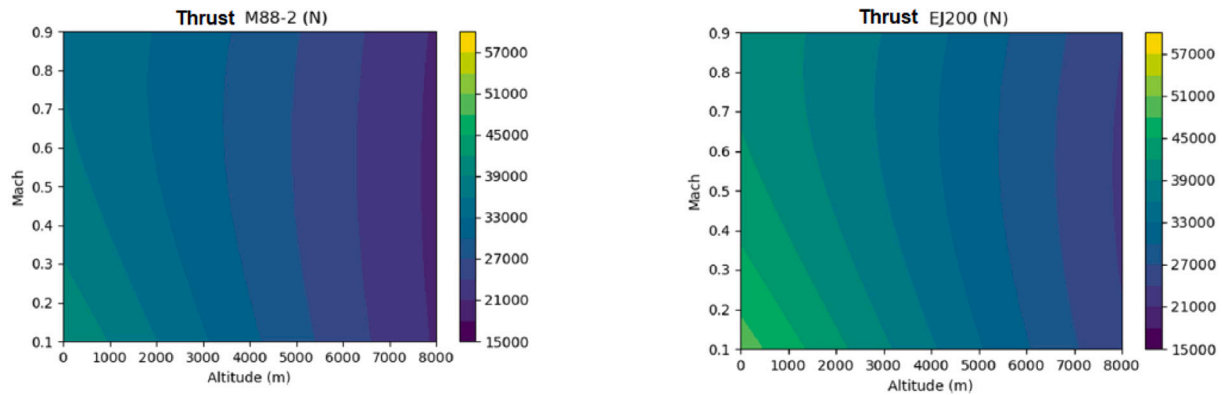


Fig. 11. Example of estimated thrust profiles as functions of the Mach number and the altitude for the M88 and EJ200 engines, using semi-empirical formulas from [35].

and a circularization burn is finally carried out at the apogee of the transfer orbit in order to inject the payload on the Sun Synchronous Orbit. During the ascent phase, the pitch and yaw angles profiles are optimized as well as the throttling of the vehicle. At the end, 13 optimization variables are involved to define the parameterized control laws (orientation angles and throttling profile). The RTLS trajectories differ for the two concepts. These latter are described in the following paragraphs.

RTLS trajectory of glide-back architecture. Once the first stage is jettisoned, the RTLS for the glide-back architecture consists of a turn around phase (using cold gas thrusters) followed by a boost-back burn (with three of the main rocket engines) to provide the impulse to go back to the landing site. This phase is controlled by two parameters: the pitch angle value of the stage during the boost-back burn and the throttling level of the rocket engines. Once the burn is achieved, the first stage carries out an atmospheric re-entry at zero degree of angle of attack. Then, when the atmospheric conditions are met, a pull-up maneuver followed by a glide phase are carried out to reach the landing site. To do so, parametric control laws with respect to the angle of attack are involved. 13 control variables are required to parameterize all the different phases of the RTLS flight. The last U-turn to be aligned to the

runway is controlled by the angle of attack as well as the bank angle. The constraints that shape the optimal RTLS trajectory of the glide-back architecture are the maximum axial load factor (for the boost-back burn), the maximal transverse load factor (for the pull-up maneuver) and the maximal admissible velocity before landing (for the last U-turn maneuver).

RTLS trajectory of fly-back architecture. The RTLS phase of the fly-back configuration begins with a full ballistic flight without any re-ignition of the rocket engines. Then, a high altitude re-entry is involved in order to satisfy the dynamic pressure and aerothermal flux constraints. The control law of this phase is parameterized by the angle of attack. Then, a U-turn maneuver is performed to align the vehicle to the runway coupled with a pull-up maneuver. Once the ignition conditions of the turbojets are met, the vehicle performs a powered cruise phase controlled by the angle of attack, the bank angle and throttling level of the turbojets. Eventually, a final U-turn phase is done to align to the runway. Piece-wise cubic monotone Hermite polynomials [37] are used as parametric laws to ensure the smoothness in the orientation angle evolution. 30 control variables are used to parameterize all the different phases of the fly-back RTLS flight. The constraints about the maximal dynamic pressure, the maximal aerothermal flux and the maximal

transverse load factor are the main drivers to shape the optimal RTLs trajectory.

3.2. Multidisciplinary design optimization process

The disciplines described in the previous sections are linked together into a Multidisciplinary Optimization Process (MDO) using the open source OpenMDAO framework developed at NASA [23]. It allows to assess the overall performance of the vehicle and to let the optimizer find the optimal compromise between the performance of the RTLs phase and the induced penalties for the ascent phase. The objective function that has to be minimized is the Gross-Lift-Off-Weight (GLOW). The decision variables are the propellant masses of the two stages, the characteristics of the reusability kit (e.g., wings and canards planforms, amount of kerosene) and the different control laws associated to the optimal control problem of the trajectory discipline. A MultiDiscipline Feasible formulation has been used with a Gauss–Seidel Fixed Point Iteration between the different disciplines [38]. For the optimization, because the minimization problem is non-linear with a high number of local optima, an evolutionary algorithm is used (Covariance Matrix Adaptation - Evolution Strategy [39]). Parametric studies have been performed on different seeds of this evolutionary algorithm to ensure its robustness to the initialization. The convergence criterion of this algorithm is the stagnation of the best solution for a defined number of iterations (typically a few dozen generations).

The overall multidisciplinary optimization problem can be formulated as follows :

$$\min \text{GLOW}(\mathbf{z}, \mathbf{y}(\mathbf{z}, \mathbf{w})) \quad (1a)$$

$$\text{w.r.t } \mathbf{z} = \{M_{p1}, M_{p2}, M_{\text{RTLs}}, \mathbf{z}_{\text{kit}}\} \quad (1b)$$

$$\mathbf{w} = \{\mathbf{w}_{\text{ascent}}, \mathbf{w}_{\text{RTLs}}\} \quad (1c)$$

$$\text{s.t. } \mathbf{g}(\mathbf{z}, \mathbf{y}(\mathbf{z}, \mathbf{w})) \leq 0 \quad (1d)$$

$$\mathbf{h}(\mathbf{z}, \mathbf{y}(\mathbf{z}, \mathbf{w})) = 0 \quad (1e)$$

$$\mathbf{g}_{\text{path}}(\mathbf{z}, \mathbf{y}(\mathbf{z}, \mathbf{w}), \mathbf{x}(t), \mathbf{w}) \leq 0 \quad (1f)$$

$$\mathbf{w}_{\min} \leq \mathbf{w} \leq \mathbf{w}_{\max} \quad (1g)$$

$$\mathbf{z}_{\min} \leq \mathbf{z} \leq \mathbf{z}_{\max} \quad (1h)$$

with GLOW the Gross-Lift-Off-Weight that has to be minimized, $\mathbf{y}$ the interdisciplinary coupling variables, $\mathbf{x}$ the state vector variables (trajectory profile), $\mathbf{z}$ the optimization variables (composed of the propellant mass of the two stages M_{p1} , M_{p2} and the propellant mass for the RTLs mission M_{RTLs} - rocket propellant for glide-back or kerosene for fly-back and $\mathbf{z}_{\text{kit}}$ the parameters of the reusability kit - e.g., wings and canards planforms). $\mathbf{w}$ stands for the parameterized control law that is composed of the control law for the ascent phase $\mathbf{w}_{\text{ascent}}$ (same for the two studied architectures) and $\mathbf{w}_{\text{RTLs}}$ the control law for the RTLs mission (differs for the glide-back and the fly-back architectures). $\mathbf{g}(\cdot)$ stands for the inequality constraints about the design whereas $\mathbf{g}_{\text{path}}(\cdot)$ represents the inequality constraint on the trajectory (e.g., maximal heat flux, maximal angle of attack, visibility). Finally, $\mathbf{h}(\cdot)$ traduces the orbit injection constraints. The resulting design process is depicted in Fig. 12.

The multidisciplinary design process presented in this section is used to carry out trade-off assessment of the two studied RLV architectures in the following section.

4. Trade-off assessment between glide-back and fly-back configurations

4.1. Glide-back architecture design

Because the glide-back concept only uses lifting surfaces in order to perform the main phase of the RTLs mission, the assessment of the optimal configuration for these aerodynamic appendices is of prime

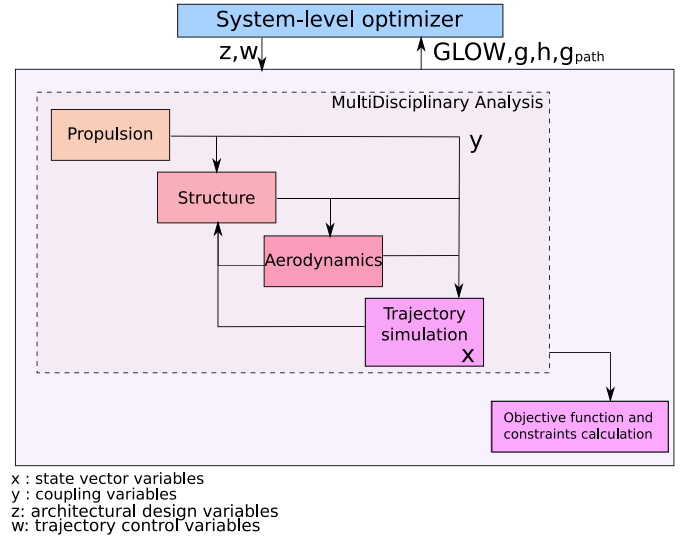


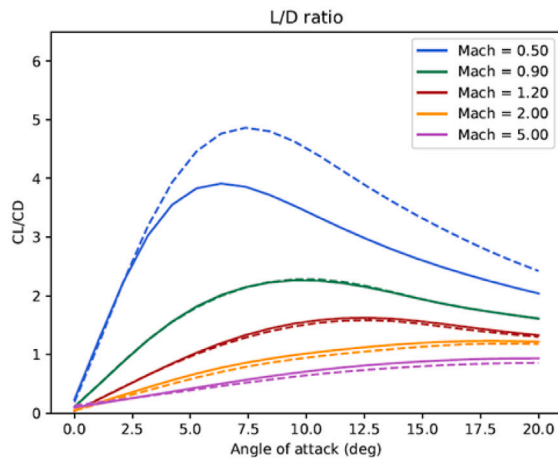
Fig. 12. Illustration of the multidisciplinary design process for the winged RLV architectures.

importance for such a vehicle. This leads to find a compromise between the maximization of the performance of the RTLs phase and the minimization of the induced aerodynamic penalties for the ascent phase. In order to select the optimal architecture with respect to the lifting surface configuration, different multidisciplinary optimizations of the glide-back concepts are performed. For each optimization, both ascent and RTLs phases are jointly taken into account, in order to be able to assess trade-offs between the different flight phases.

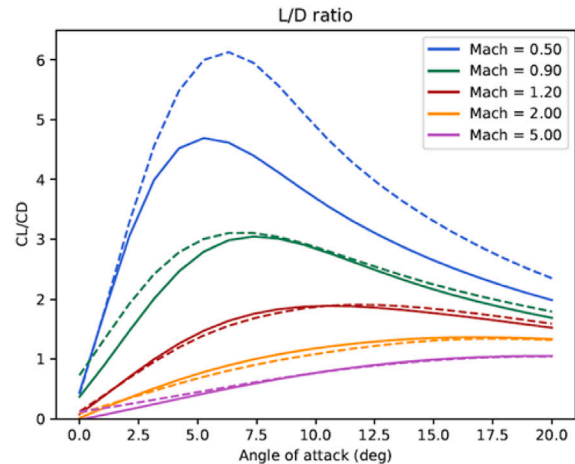
The main design drivers for lifting surfaces are the planform and the airfoil. In this study, two types of planform and two types of airfoil have been considered. In order to comply with the reusability kit constraints, the chord of the wing is set to the thrust frame length. In the same way, the trailing edge sweep angle is set to 90 degrees to comply with launchpad constraints. Considering the planform, two wing spans have been studied, respectively to 13.8 m (equivalent to three times the stage diameter) and 18.4 m (equivalent to 4 times the stage diameter). Wingspans are constrained by the launch pad facilities and therefore larger wingspans have not been considered in this study. Considering the airfoil, two types of airfoil have also been analyzed. The choice of the candidate airfoil is made according to their performance in different Mach number regimes. Indeed, double-wedge airfoil (which is efficient in supersonic flight regime) is compared to traditional NACA airfoil (efficient in subsonic flight, but leading to increased drag in supersonic regime).

Fig. 13 summarizes the estimated the lift over drag ratios for the different aerodynamic configurations using the aerodynamic modeling discussed in Section 3.1.2. As it can be seen, NACA airfoil provides a better lift over drag ratio about one point for a Mach number of 0.5 with respect to the double-wedge airfoil. The relative gain is approximately the same between the two candidate wingspans. Even if the NACA airfoil and large wingspan seem to provide better aerodynamic performance for the RTLs phase, these two characteristics lead to an increase of drag in the ascent phase and consequently global trade-off needs to be assessed.

In order to perform the lifting surface trade-off on the overall design, three configurations have been optimized using the multidisciplinary process presented in Section 3.2, respectively double-wedge airfoil and large wingspan (named Glide-back double-Wedge Large wingspan, GWL), NACA airfoil and small wingspan (named Glide-back NACA Small wingspan, GNS), and NACA airfoil with large wingspan (named Glide-back NACA Large wingspan, GNL). The corresponding optimal trajectories are depicted in Fig. 14.



(a) Aerodynamic comparative analysis with respect to the airfoil type for a wingspan of 6.9m: in solid line : double-wedge airfoil, in dash line : NACA airfoil



(b) Aerodynamic comparative analysis with respect to the wingspan for NACA airfoil: in solid line : 13.8m, in dash line : 18.4m

Fig. 13. Evolution of the lift over drag ratio with respect to the angle of attack and the Mach number.

Table 2
Mass breakdown for the different optimal architectures.

		GWL	GNS	GNL
	GLOW (t)	495.6	490.2	483.1
First stage	Propellant mass for ascent (t)	303.2	299.6	293.4
	Propellant mass for RTLS (t)	26.3	25.9	23.5
Second stage	Propellant mass (t)	110.7	109.9	111.0

Table 3
Propellant mass breakdown for the RTLS boost-back burn.

	GWL	GNS	GNL
Overall propellant mass for the RTLS phase (t)	26.4	25.9	23.5
Propellant mass to cancel the tangential velocity (t)	23.6	23.2	22.3
Propellant mass for the RTLS impulse (t)	2.8	2.7	1.2

The ascent trajectories are quite the same for the three configurations, in terms of both altitude and velocity profiles (Fig. 14). This can be explained by a same order of magnitude of drag penalties during the ascent phase for the different configurations. One has to mention that the main part of the ascent of the first stage is performed at zero degree for the angle of attack to minimize the aerodynamic loads. However, the RTLS phase profiles are quite different. Indeed, the altitude profile with respect to the distance to the landing site shows a better efficiency of the GNL architecture with respect to the two others. It can be explained by its better lift over drag ratio as showed in Fig. 13.

The propellant masses for the different optimized architectures are summarized in Table 2. As it can be seen, the best architecture is the GNL in terms of GLOW. Indeed, this architecture uses the less propellant mass for the RTLS mission (23.5t) that induces less penalty for the ascent phase of the first stage (293.4t of propellant are required whereas GNS requires 299.6t and GWL requires 303.2t). Furthermore, the staging point between the three architectures is quite analog. Indeed, the discrepancy of the mass propellant required for the second stage is about 1 ton (less than 1%).

It also can be valuable to analyze in depth the amount of propellant required for each phase of the RTLS boost-back burn. Typically the boost-back burn can be divided in two phases (Fig. 15): the first is responsible for the cancellation of the tangential velocity and the second is responsible for providing the impulse to the landing site. As it is summarized in Table 3, the mass required for the second phase of the boost-back burn is very reduced for the GNL configuration with

Table 4
Mass breakdown for the reusability kit of the glide-back configuration corresponding to Fig. 3.

Lifting surfaces and rear case	2.3t
Nose and front skirt	1.5t
Gears	2.0t
Propellant	23.5t
Global mass of reusability kit	29.3t

respect to the two others. This can be explained by the increased lift for this configuration that enables a greater distance in gliding mode (73 km for the GNL with respect to about 55 km for the two other configurations, see Fig. 14). Indeed, the gliding phase is mainly subsonic (around a Mach number of 0.6) and therefore the NACA profile provides substantial aerodynamical advantages compared to the double-wedge airfoil and consequently lead to a smaller impulse boost need for the RTLS.

Consequently, for the trade-off assessment concerning the glide-back architecture, it can be seen that the advantages of both subsonic airfoil and larger wingspan is predominant with respect to the relative losses during the ascent phase. The GNL configuration is therefore selected as the best glide-back architecture for the overall trade-off. The reliability kit mass breakdown for the GNL configuration is detailed in Table 4. The main part of the reusability kit mass corresponds to the additional propellant mass for the boost-back burn.

4.2. Fly-back concept design

The technological choices for the fly-back solutions are quite larger than for the glide-back concepts. Indeed, in addition to the lifting surface, these types of concepts also use additional air-breathing propulsion system for the RTLS mission. In this study, two military engines are considered as candidates for the RTLS mission. These are the Snecma M88 [40] and the Eurojet EJ200 [41] (Table 5). These engines have

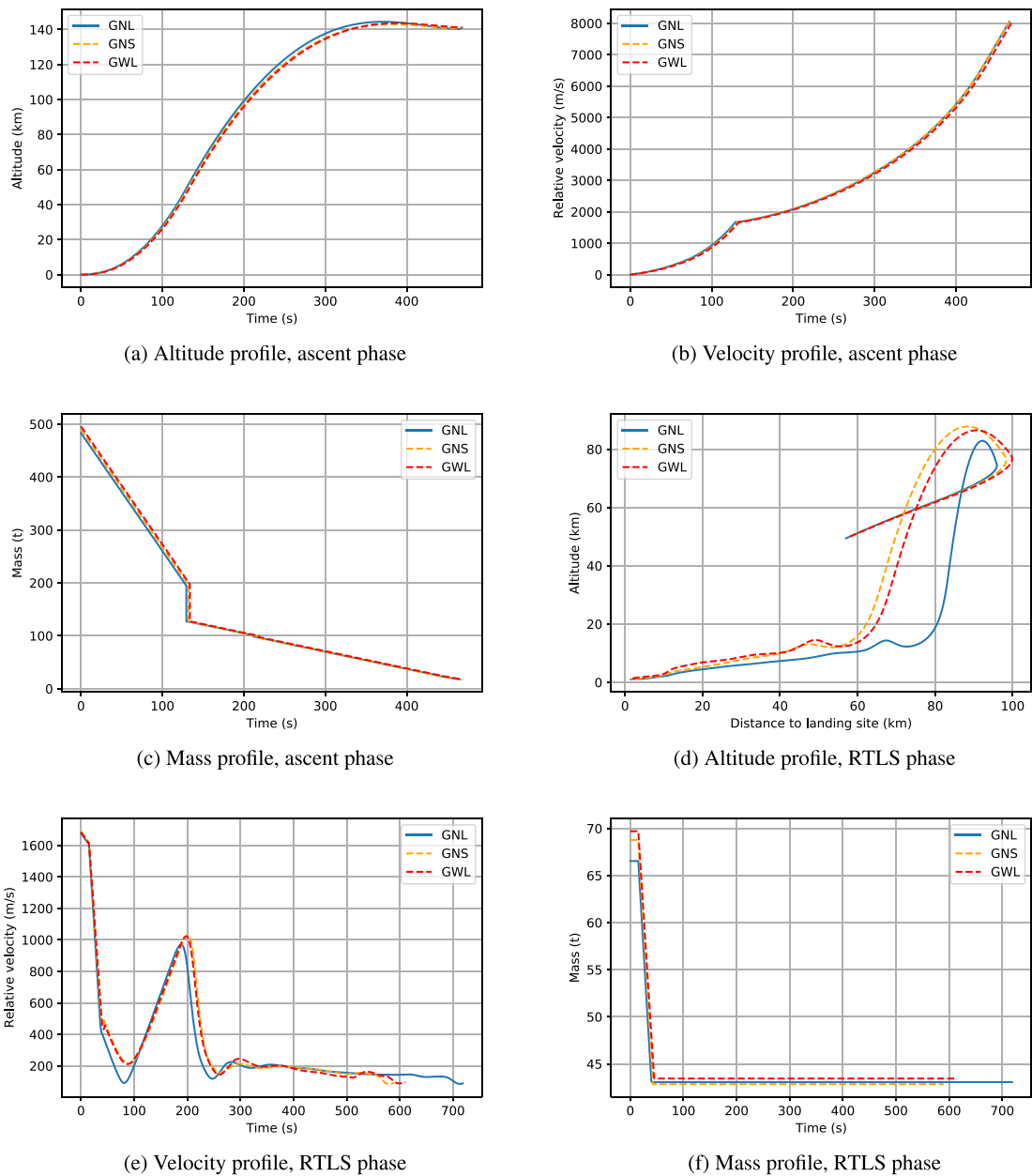


Fig. 14. Comparative analysis of optimal trajectories for GNL, GNS and GWL architectures.

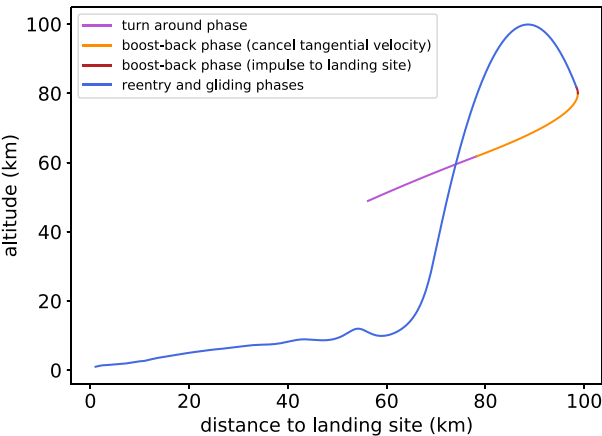


Fig. 15. Details of boost-back burn.

Table 5
Characteristics of turbojet engines (manufacturer data [40,41]).

	EJ200	M88
Thrust at ground (kN)	60	50
Mass (kg)	990	897
Length (m)	4	3.54
Diameter (m)	0.74	0.69
Specific fuel consumption (cruise) (10^5 (kg/s)/N)	2.1	2.2

been selected with respect to their unitary thrust and the compactness of the engine systems. These engines have low by-pass ratio that is consistent with the integration into the reusability kit.

Considering the turbojet engine impact, two main analyses are carried out regarding the influence of the number of turbojet engines (using EJ200 engines) and an analysis regarding the influence of the

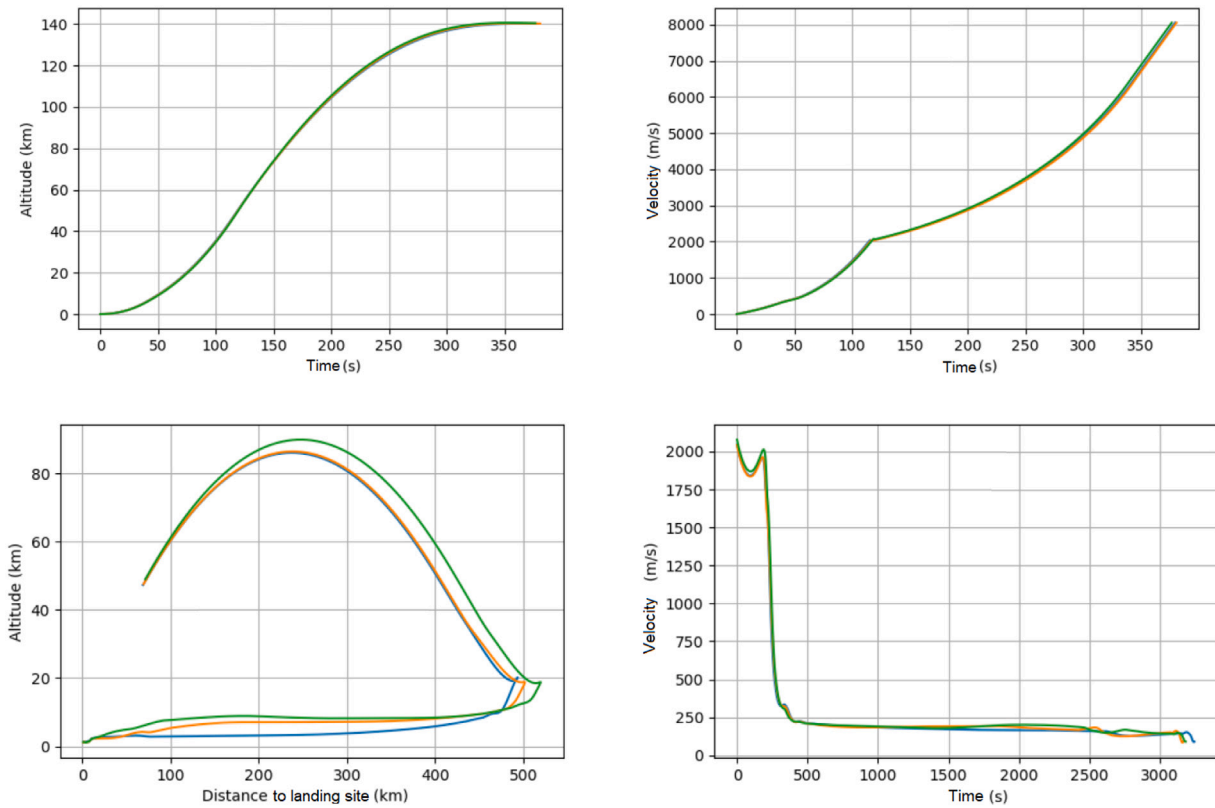


Fig. 16. Ascent phase profile (top left altitude, top right velocity) and RTLS mission (bottom left altitude, bottom right velocity) for fly-back architectures with 2 engines (blue), 3 engines (orange) and 4 engines (green). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

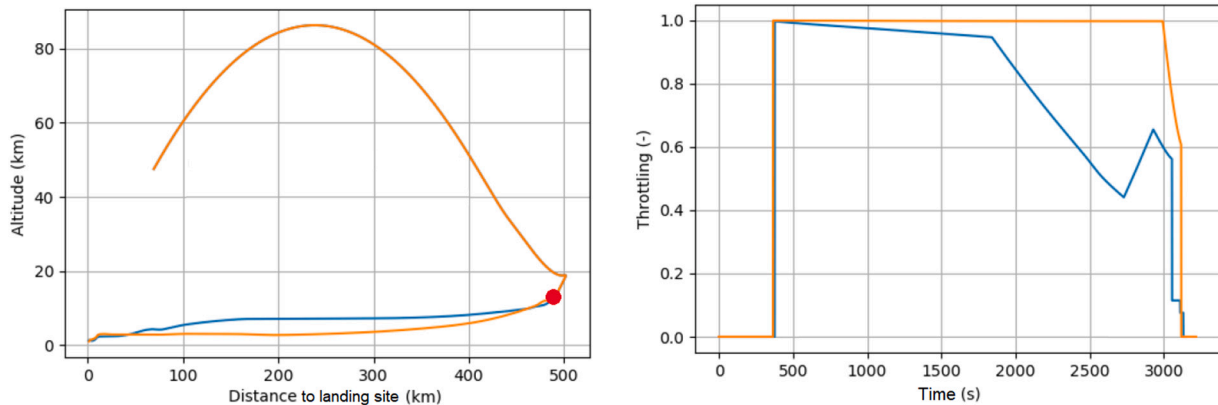


Fig. 17. Abort scenario, left: altitude profile as a function of the distance to the landing site, right: throttling of the turbojets as a function of the time. In blue: nominal trajectory, in orange: abort trajectory, in red: failure point (loss of one turbojet). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

type of engines between EJ200 and M88. In the following section are described the main studies to assess optimal trade-offs for the fly-back architecture.

4.2.1. Impact of the number of turbojet engines

Nominal mission. In order to assess the trade-off concerning the number of turbojets that have to be included in the reusability kit, three configurations using EJ200 turbojets are studied: architectures with 2, 3 and 4 turbojet engines. The number of turbojets has an influence on the conditions of the cruise flight. With more thrust, the RTLS cruise flight has more flexibility to be performed at a higher altitude limiting additional drag and consequently limiting the required kerosene, at the price of complexity of integration and more dry mass (additional engines) that is a direct loss for the ascent phase. The different optimal

trajectories are described in Fig. 16. The aerodynamic configuration of the fly-back architecture is the same as the GNL (NACA airfoil, large wingspan).

While the ascent phase is similar for the different architectures (the configurations mainly differ in terms of total turbojet engine masses and kerosene mass for RTLS), the RTLS phase is quite different, both in terms of the altitude of the cruise flight and the maximal distance to landing site after the atmospheric reentry. Indeed, the configuration with 4 engines (green) has the maximal distance to return to the landing site. Moreover, as expected, the higher the thrust, the higher the cruise altitude (9 km for 4 engines, 7 km for 3 engines and 3.2 km for 2 engines). Considering the Gross-Lift-Off-Weight for this configuration, both the configurations with 2 and 3 engines provide quite similar results (few tons of difference) whereas the architecture with 4 engines

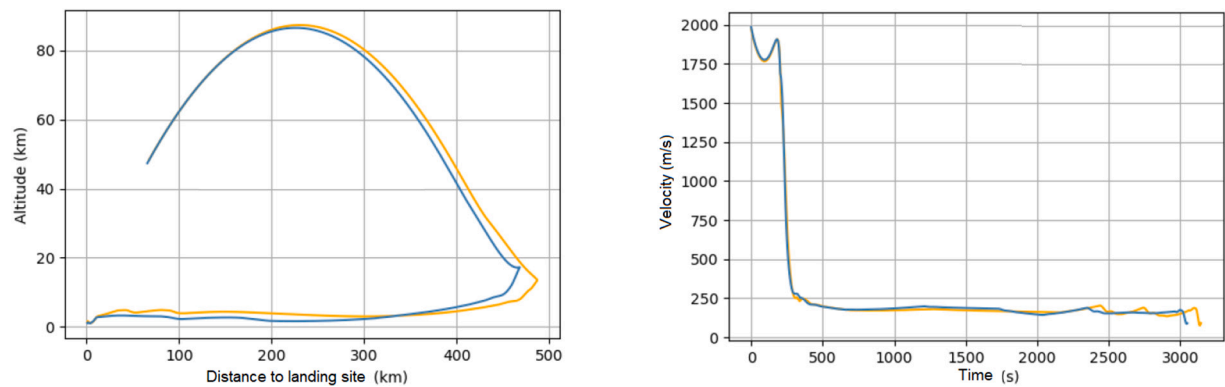


Fig. 18. Return trajectories for different turbojet types, at left: altitude profile as a function of the distance to landing site, at right: velocity as a function of time. In blue: RLV architecture with three M88, in orange: RLV architecture with three EJ200. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

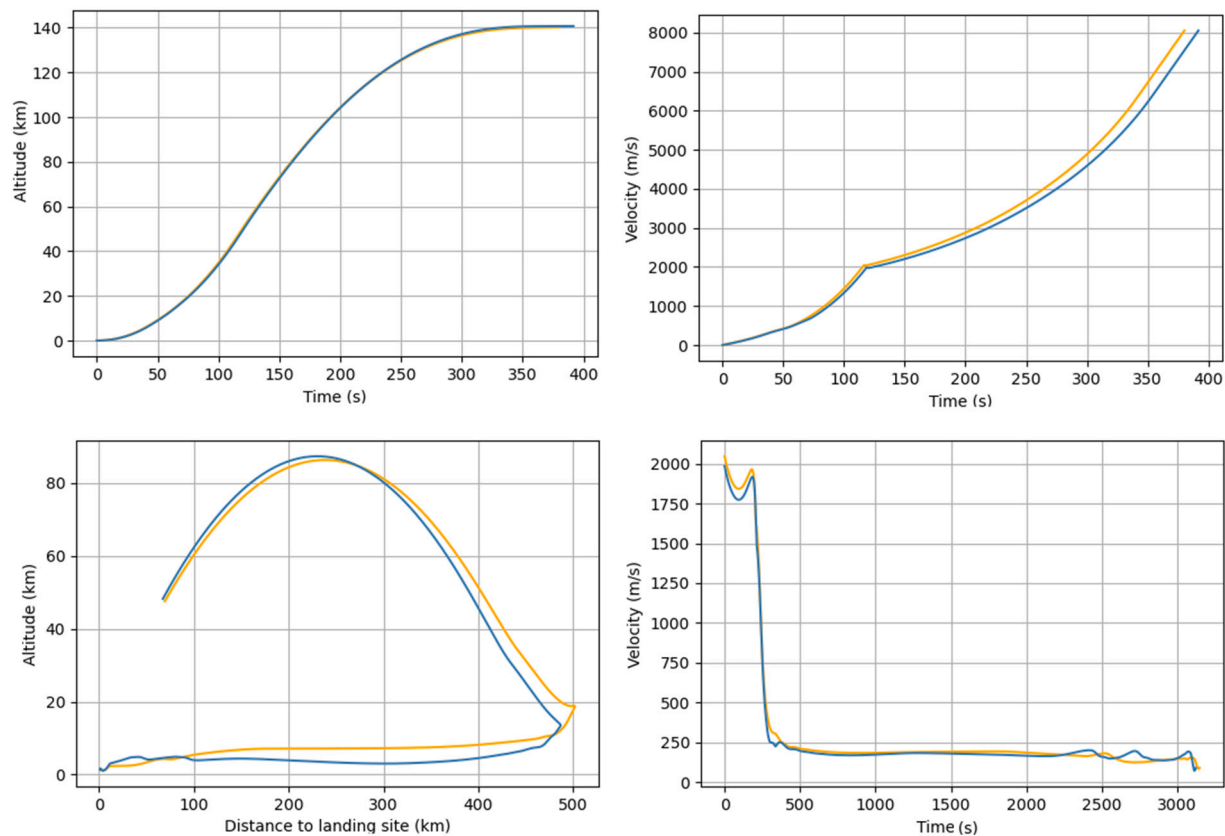


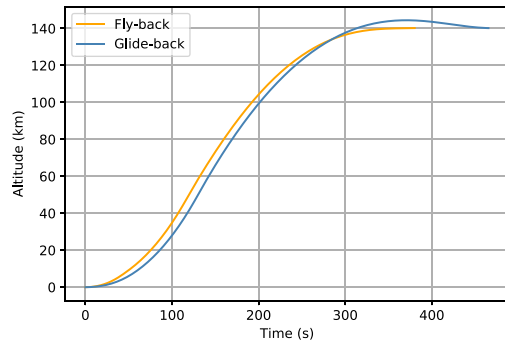
Fig. 19. At the top: ascent trajectory profiles. At the bottom: RTLS mission, with in blue: small wingspan, and in orange: large wingspan. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 6		2 engines	3 engines	4 engines
Mass breakdown for the different optimal architectures.				
First stage	GLOW (t)	407.4	410.5	415.5
	Propellant mass for ascent (t)	259.6	261.8	267.3
	Kerosene mass for RTLS (t)	5.0	5.0	5.5
Second stage	Propellant mass (t)	86.0	85.9	84.0

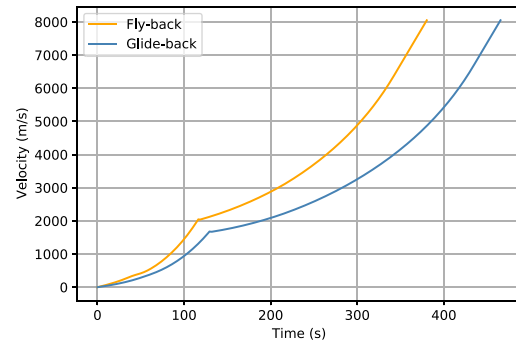
is 8 tons heavier (comprising 500 kg of additional kerosene with respect to the other configurations), with a baseline about 410.5 tons (Table 6).

Robustness with respect to engine failure. As the configurations with 2 and 3 engines are very close in terms of performance (GLOW), an

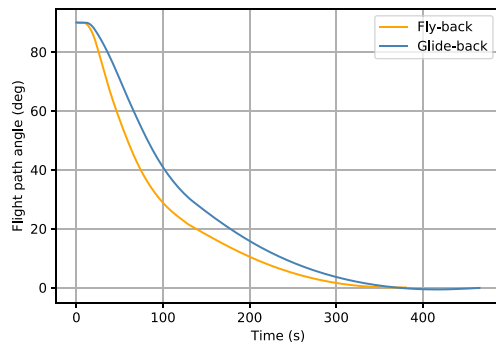
additional study is conducted in order to identify whether the configuration with 3 engines could provide more robustness with respect to a turbojet engine failure. For that purpose, a simulation of a specific scenario is carried out considering that only 2 of the 3 engines are ignited at the beginning of the cruise flight. To do so, the control



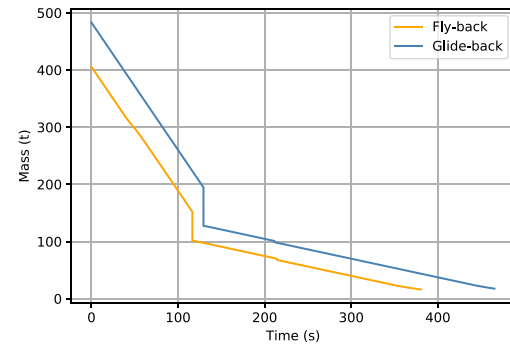
(a) Altitude as a function of time for ascent phase



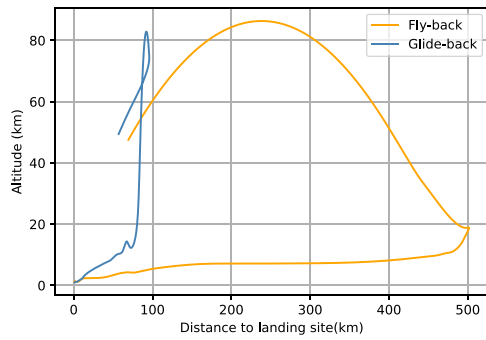
(b) Relative velocity as a function of time for ascent phase



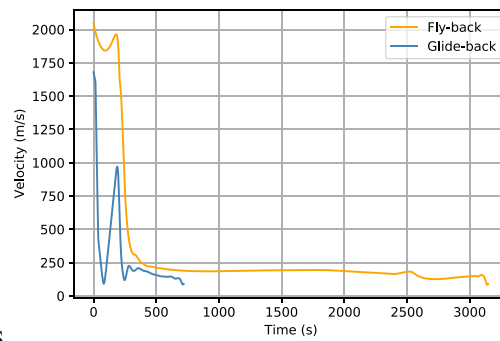
(c) Flight path angle as a function of time for ascent phase



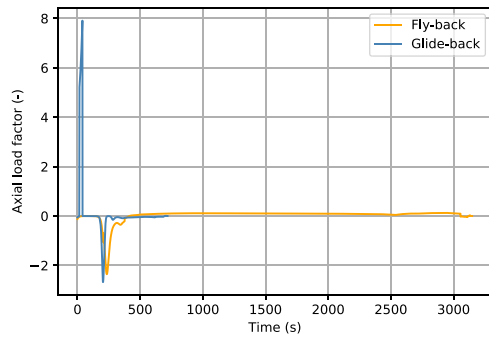
(d) Mass as a function of time for ascent phase

Fig. 20. Ascent phase profiles for fly-back and glide-back architectures.

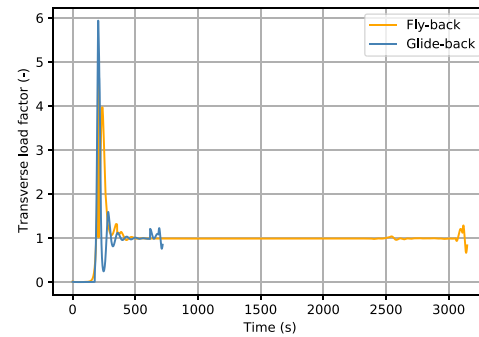
(a) Altitude as a function of distance to landing site for RTLS phase



(b) Relative velocity as a function of time for RTLS phase



(c) Axial load factor as a function of time for RTLS phase



(d) Transverse load factor as a function of time for RTLS phase

Fig. 21. RTLS phase profiles for fly-back and glide-back architectures.

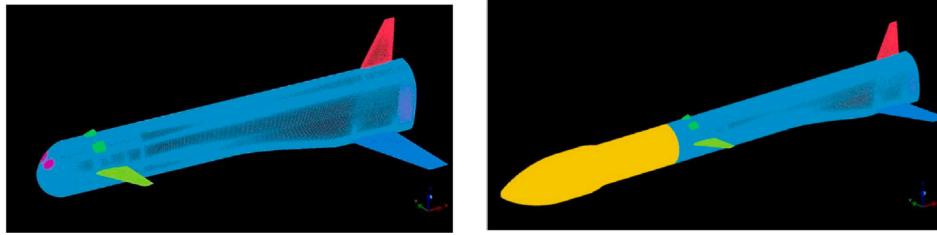


Fig. 22. Aerodynamical meshes for composite and first stage configurations.

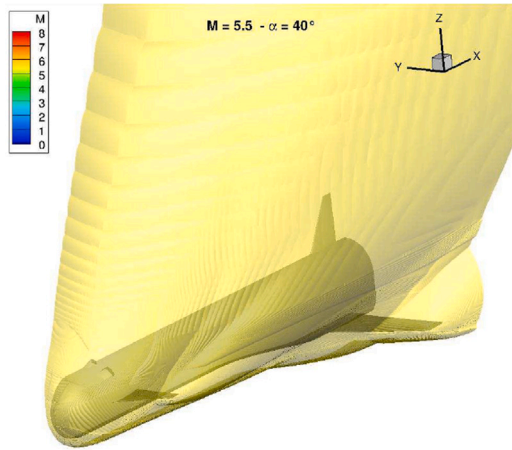


Fig. 23. RANS calculation of reentry phase.

law for this flight phase (profile of angle of attack and bank angle, throttling of turbojets) are reoptimized considering the loss of one engine. The resulting RTLS trajectory is described in Fig. 17. As it can be seen on this figure, the trajectory with one turbojet out of order remains feasible. Consequently, the architecture with 3 turbojets seems to present the best trade-off between the reliability of the configuration (redundancy of one turbojet) and performance (GLOW).

4.2.2. Impact of the engine type

In order to assess the trade-off considering the type of engine, both configurations with three M88 and three EJ200 are designed. For this technological choice, the trade-off involves the compactness of the propulsive system, for the sake of integration into the reusability kit, and the overall performance of the reusable launch vehicle. Both architectures with three M88 or three EJ200 are designed using the MDO process presented in Section 3.2. The resulting trajectories are described in Fig. 18.

The results of the two optimizations show that both configurations are quite close in terms of GLOW (the configuration with three M88 is 1.5 tons lighter than the other). This is due to the fact that the specific fuel consumption of the M88 is a little bit lower than the one of the EJ200. It allows to save around 100 kg of kerosene for the return mission, that leads to 1.5 tons of savings considering the impact on the ascent flight. Furthermore, considering the compactness of the two types of engines, the EJ200 presents advantages in terms of length, diameter and unitary mass (Table 5). Consequently, from the two characteristics (performance and integration of propulsive system), it is chosen to select the EJ200 engine for the following of the study.

4.2.3. Impact of wing planform

The same two wing planforms as for the glide-back are considered in the fly-back architecture (see Fig. 4). Indeed, a larger wingspan can present advantages for the RTLS cruise flight by delivering more

lift at the price of additional drag for the ascent phase. This can be even more valuable as for the fly-back concept, a long cruise phase has to be performed. Consequently, a global trade-off needs to be assessed. Only the NACA airfoil has been studied due to the cruise flight regime of the fly-back concept (subsonic flight). To evaluate the trade-off, configurations with three EJ200 and the different wingspans is designed. As it can be seen on Fig. 19, the wingspan has a great effect on the return trajectory but also on the ascent velocity profile. Indeed, the different wingspans cause different drag effects that can be noticed in the ascent velocity profiles. On the performance point of view, the configuration with the longer wingspan presents better characteristics in terms of GLOW (7 tons of difference). Indeed, the smaller wingspan induces a reduction of about 1 point in terms of lift-over-drag ratio (at cruise conditions), that causes 1.6 tons of additional kerosene to be put on the reusability kit to perform the RTLS mission. Consequently, this mass induces additional rocket propellant for the ascent phase that in overall generates a difference of 7 tons on the global launch vehicle. Consequently, it appears that the largest wingspan configuration is more suitable for the RTLS mission. This configuration has been selected for the rest of the study.

4.2.4. Synthesis of the fly-back architecture trade-off

The different trade-off studies described in this section involved the choice of the engine type, the choice of the engines number and the wing configuration. To assess the trade-offs, multiple multidisciplinary design optimizations have been performed in order to assess the compromises between the ascent and return phases. Furthermore, several other aspects such as the redundancy or the engine integration have also been studied. From all these investigations, the configuration with three EJ200 turbojet engines and the larger wingspan configuration is selected as providing the best compromises between the overall performance (GLOW), the reliability of the RTLS mission (the trajectory with the loss of one engine is feasible) and the engine integration (EJ200 is more compact than M88).

In order to analyze more in-depth this configuration, several model refinements are made to address specific keypoints of the design. These refinements are described in Section 5.

4.3. Comparative analysis between glide-back and fly-back configurations

In this section, the glide-back and the fly-back architectures are compared in order to identify which one is the most promising with respect to the mission specifications.

In Table 7, the best architectures for fly-back and glide-back return are compared. As it is summarized, the fly-back architecture is more efficient than the glide-back concept in terms of GLOW. Indeed, its GLOW is lighter by 72.6t. Whereas the glide-back first stage is penalized by the propellant mass for the RTLS mission (23.5t), the fly-back benefits for the efficiency of the turbojets to present less consumption of kerosene for this phase. Consequently, the total mass of the first stage is about 307.8t for the fly-back architecture and 355.0t for the glide-back architecture. Furthermore, this global optimization of both the ascent and RTLS phases allows to find optimal compromises between the two

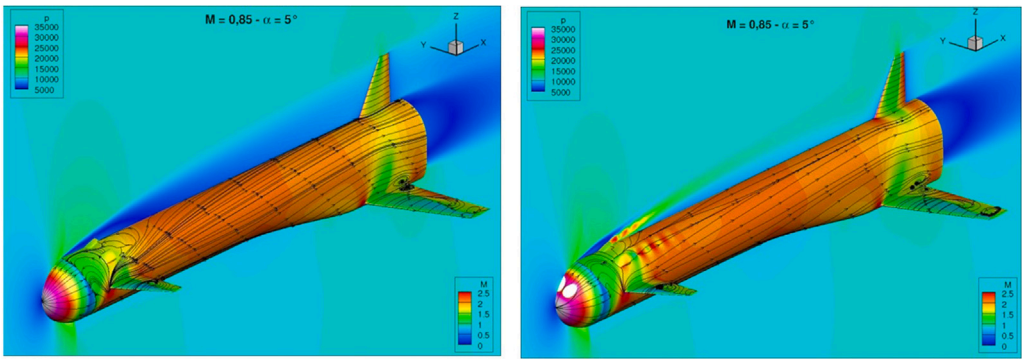


Fig. 24. RANS calculations of RTLS cruise flight with at left: turbojets off and at right: turbojets on.

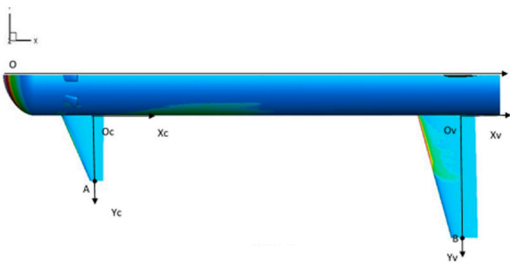


Fig. 25. Handling of aerodynamic loads into the structural analysis.

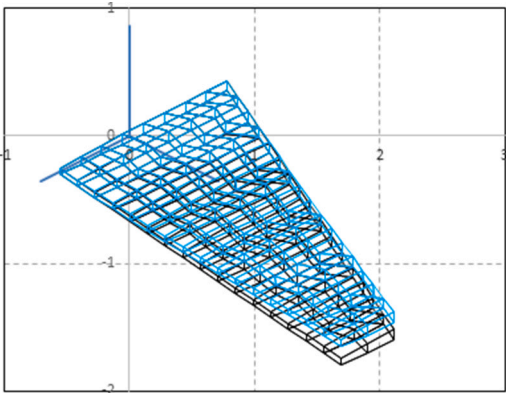


Fig. 26. Calculation of wing deformation during cruise flight.

Table 7
Characteristics of fly-back and glide-back architectures.

		Fly-back	Glide-back
First stage	Propellant mass (t)	261.8	293.4
	Dry mass (t)	40.9	38.1
	Propellant mass RTLS (t)	5.0	23.5
	Total mass (t)	307.8	355.0
Second stage	Propellant mass (t)	85.9	111.0
	Dry mass (t)	7.1	7.9
	Total mass (t)	93.0	118.9
Upper composite and interstage (t)		9.7	9.2
GLOW (t)		410.5	483.1

phases and leads to different staging points for the two architectures (and consequently mass of the expendable second stage).

In Table 8, the staging points (*i.e.*, the separation between the first and second stages) of the two proposed architectures are compared. Although the altitudes of the staging points are quite comparable, the velocity at separation for the fly-back architecture is higher than the

Table 8
Characteristics of staging point for fly-back and glide-back architectures.

	Fly-back	Glide-back
Altitude (km)	47.6	49.3
Velocity (m/s)	2046	1679
Mach number (-)	6.2	5.1
Distance to landing site (km)	69.4	57.0
Flight path angle (deg)	24.3	30.5

one of the glide-back architecture, with a discrepancy around one Mach number. This is due to the fact that for the fly-back architecture, there is no boost-back burn at the difference of the glide-back concept. For the glide-back, a higher Mach number at the staging condition leads to an harder cancellation of the horizontal velocity with the boost-back and therefore a larger propellant mass consumption. Furthermore, the ascent trajectory of the fly-back concept is less vertical than for the glide-back architecture. Indeed, the distance to landing site at separation is about 69.4 km for the fly-back architecture and 57.0 km for the glide-back architecture. In the same way, the flight path angle at separation for the fly-back concept is by 6 degrees smaller with respect to the glide-back concept. This can also be explained by the boost-back maneuver involved in the glide-back architecture. Indeed, this maneuver is used to mainly cancel the horizontal velocity of the first stage to come back to the landing site. If the trajectory is more vertical (that can be translated by a larger flight path angle value at the separation), the horizontal velocity is lower and consequently the propellant that has to be carried out for the RTLS mission is also more limited. This relative vertical trajectory induces several constraints on the ascent phase that result in a lower optimal GLOW.

The trajectory of the two configurations are summarized in Fig. 20 for the ascent phase and Fig. 21 for the RTLS phase. As it can be seen, the two ascent trajectories are quite identical. Because the first stage of glide-back architecture has to carry out a larger mass of propellant for RTLS, the acceleration of this concept is lower than the fly-back architecture. The main differences between the two architectures are visible in the RTLS profiles. Indeed, one can see that the two architectures present very different trajectory profiles for the RTLS mission. In terms of the altitude as a function of the distance to landing site, the glide-back architecture gliding phase is quite small with respect to the fly-back cruise (about 500 km). Moreover, the loads undertaken by the glide-back concept are greater than the fly-back concept. Indeed, because the glide-back architecture uses the main rocket propulsion, the axial load factor at the beginning of the RTLS flight is quite large (around 8.0) whereas the fly-back does not see critical loads in this phase. Furthermore, the pick of transverse load factor is undertaken during the pull-up maneuver for the two vehicles. Once again, the glide-back architecture endures higher loads than the fly-back concept due to the fact that this architecture has to perform full gliding RTLS phase. These higher loads involve penalizations in terms of dry mass

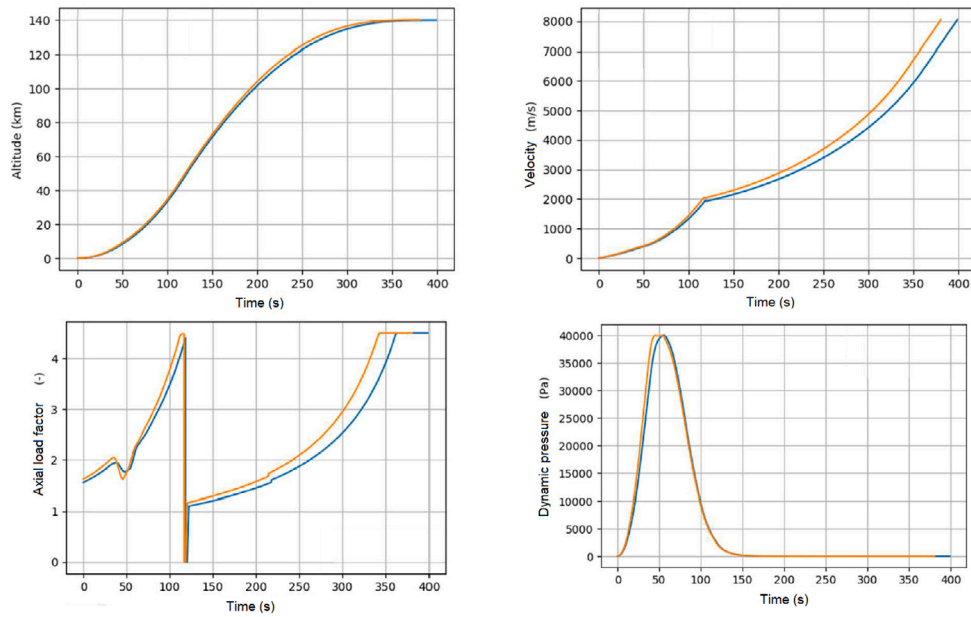


Fig. 27. Ascent phase trajectory profiles, with in orange: the reference trajectory and in blue: the trajectory obtained with refined models. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

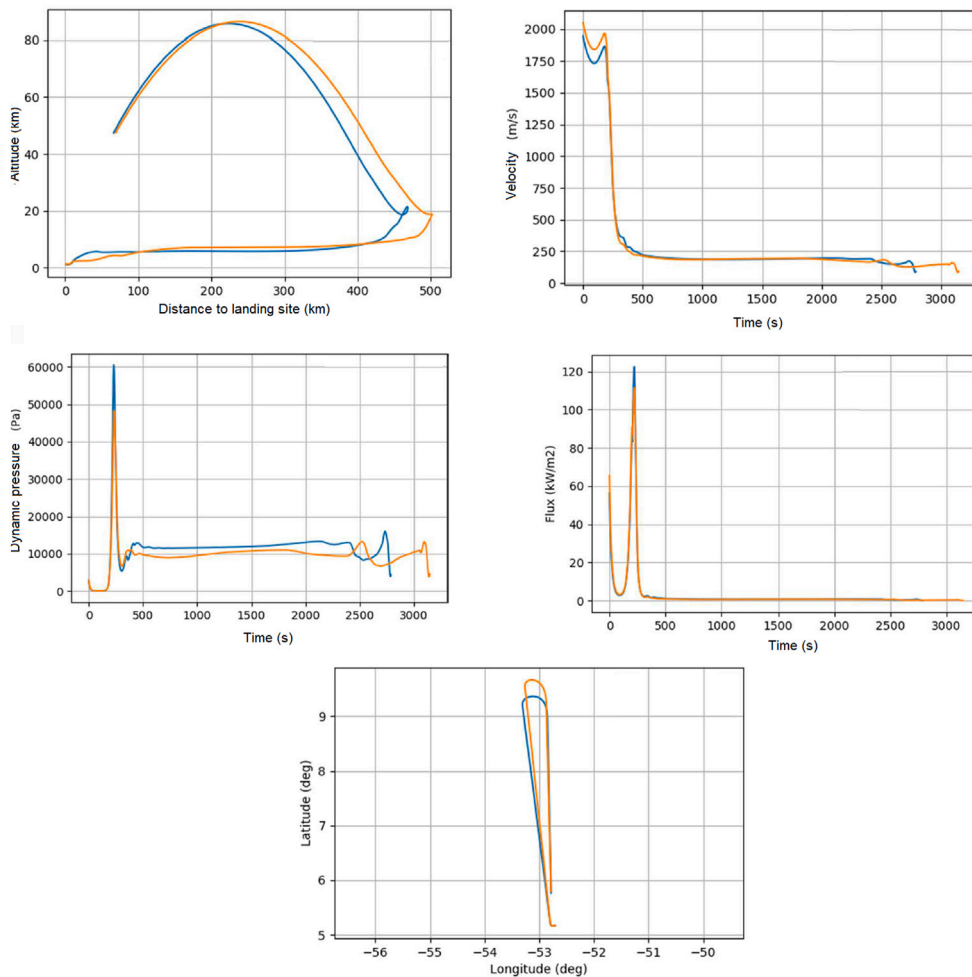


Fig. 28. RTLS trajectory profiles, with in orange: the reference trajectory and in blue: the trajectory obtained with refined models. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

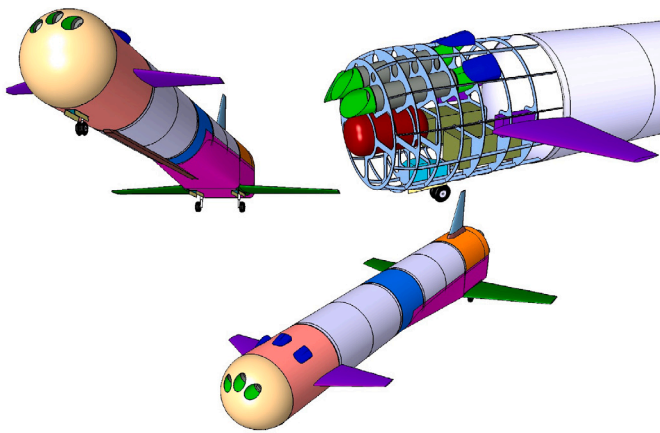


Fig. 29. Illustrations of the first stage with the reusability kit.

for this concept. Eventually, one can see that the cruise flight of the fly-back concept is longer (around 2500 s) with respect to the glide-back concept, and this phase is performed at a cruise subsonic flight (Mach number around 0.6).

Considering these trade-off studies and the interest of the fly-back architecture, a refinement of disciplinary models is carried out in the following Sections to further analyze such an architecture and to improve the accuracy of the performance estimation.

5. Additional studies on the fly-back configuration

5.1. Aerodynamics

Concerning the aerodynamics performance of the launch vehicle, RANS (Reynolds Averaged Navier Stokes) high-fidelity calculations have been performed on several key flight points (around sixty RANS calculations) (Fig. 22).

The first critical key-point concerns the reentry at a high angle of attack. The CFD calculations allow to evaluate the position of the shock at different flight configurations (in terms of altitude, angle of attack and Mach number) to evaluate the feasibility of the reentry (Fig. 23).

The second critical key-point is the perturbation of the flow due to the ignition of the turbojets. To evaluate this aspect, different RANS calculations have been performed with turbojets on and off in order to identify the impact in terms of lift decrease (Fig. 24) and to refine the aerodynamics coefficients estimation for the powered RTLS cruise flight.

5.2. Air-breathing propulsion

The air-breathing engine performance have also been refined by using an in-house thermodynamic cycle simulation code in order to simulate the thrust and specific fuel consumption on the obtained reference trajectory. To do so, an off-design phase has been conducted to simulate the EJ200 engine characteristics. Then, an on-design study has been carried out to evaluate the performance of the engine and to compare with the preliminary results obtained in the trade-off assessment phase. Whereas the thrust profile directly corresponds to the one used in the preliminary phase, the specific fuel consumption differs by 10%. This discrepancy has been included for refining the performance of the vehicle consumption.

5.3. Structural analysis

The structural analysis takes into account all the aerodynamics loads provided by the refined aerodynamics assessment that has been derived. This allows to consider the lift distribution along the wings, fuselage and canards (Fig. 25) as well as the pitching moment and bending forces especially during the reentry and pull-up maneuver (Fig. 26). From the Euler and RANS calculations, aerodynamic loads are extracted and used for the sizing of the lifting surfaces and the determination of the internal structures (e.g., ribs and spars) (Fig. 26). Concerning the main core, in the ascent phase, the tanks are pressurized and subjected to compression (rocket engine thrust) as well as the skirts (non pressurized). The aerodynamic loads on the lifting surfaces are not large due to the use of a symmetric airfoil and a flight at zero degree of angle of attack during the gravity turn. During the RTLS, the aerodynamic loads are not negligible. In this phase, the main core is submitted to bending moments induced by the mass of the launch vehicle, the load factors and the aerodynamic loads induced by the lifting surfaces. The wings and canards are located at the top and bottom of the main core that increase the bending moment of the tanks. Consequently, these loads have also to be taken into account in the refinement of the structural analysis of the tanks using finite element analysis. With these refinements, the reusability kit mass estimation is more accurate as it directly accounts for the aerodynamic loads.

5.4. Performance assessment

The refined disciplinary models are reintegrated into the multi-disciplinary design process (Section 3.2) taking into account the new aerodynamic coefficients estimations, the new mass model and the new air-breathing engine data. The fly-back concept is updated considering this new multidisciplinary design process by running a new optimization problem. The updated optimal trajectories are described in Figs. 27 and 28. Differences can be noted on these trajectories, both for the ascent phase (especially on the velocity profile as seen on the pitch angle profile) and the return phase (especially concerning the maximal distance to the landing site). Indeed, the refinement of the structural model (Table 9) leads to a higher structural mass for the same configuration (40.9t for the former fly-back concept and 46.9t for the updated fly-back) and therefore a higher overall GLOW (410.5t for the former fly-back compared to 423.4t for the updated one) and propellant masses for the ascent phase (261.8t compared to 263.5t) giving in the end a lower velocity condition at the stage separation for the new concept (Fig. 27).

Regarding the RTLS, the distance from landing site after the pull-up maneuver is smaller with the refined models from about 50 km due to a velocity at staging condition that is lower, resulting in a lower need for propellant mass of kerosene (4.7t instead of 5.0t for the former fly-back model). The loads undertaken during the RTLS are in the same order of magnitude and respect the allowed maximal loads. The cruising conditions for the RTLS are similar in terms of altitude and velocity. Consequently, considering both the ascent phase and the RTLS, the global performance using refined models has grown of approximately 5% in terms of GLOW mainly due to an increase in the dry mass of the launcher. The first stage of the final fly-back architecture with updated performance is illustrated in Fig. 29.

5.5. Robustness analysis

In order to consolidate the design and to refine the design margins (given by CNES), a robustness analysis has been performed with respect to model uncertainties (e.g., rocket engine specific impulse, subsystems masses, drag coefficient estimation). Because this type of architectures has never flown, it is quite important to evaluate the dispersion of the performance with respect to model uncertainties. For that purpose, in collaboration with CNES, eight uncertain parameters are considered:

Table 9
Characteristics of fly-back architectures (with updated models).

		Fly-back	Updated Fly-back
First stage	Propellant mass (t)	261.8	263.5
	Dry mass (t)	40.9	46.9
	Propellant mass RTLS (t)	5.0	4.7
	Total mass (t)	307.8	315.2
Second stage	Propellant mass (t)	85.9	91.7
	Dry mass (t)	7.1	7.3
	Total mass (t)	93.0	99.0
Upper composite and interstage (t)		9.7	9.2
GLOW (t)		410.5	423.4

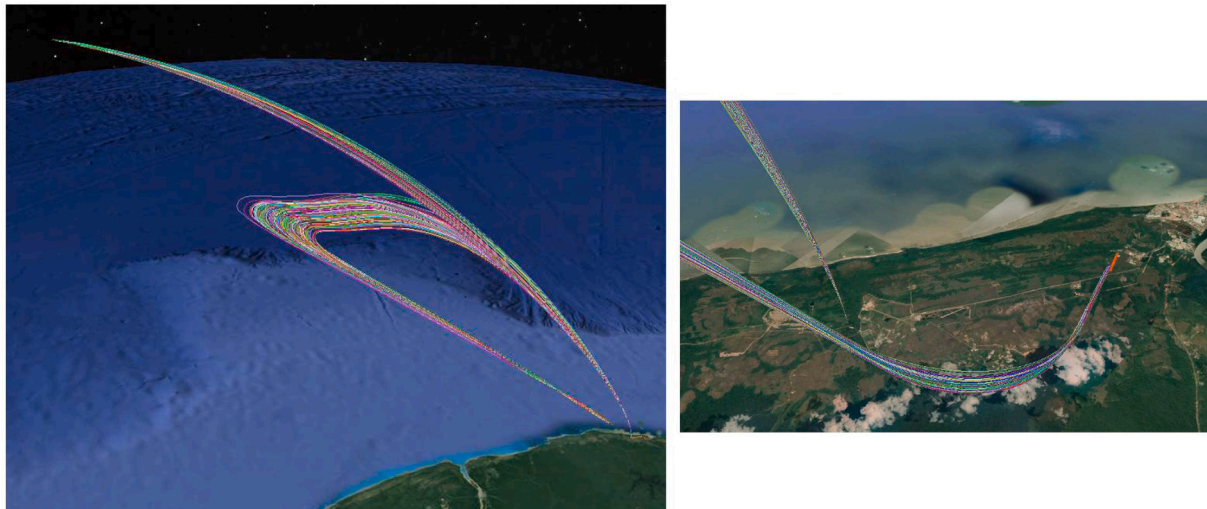


Fig. 30. Dispersion of trajectories for both ascent and return phases.

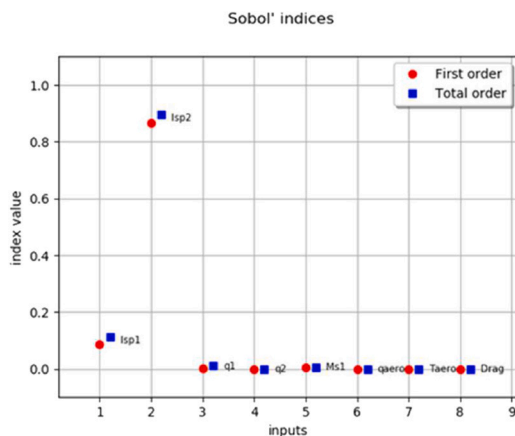


Fig. 31. Sensitivity analysis on the overall performance for the different uncertainties (Isp: specific impulse, q: mass flow rate, Ms: dry mass, T: thrust).

on the specific impulse of the PROMETHEUS engine for both the first and the second stages, on the mass flow rate of the rocket engines, on the thrust and specific fuel consumption of the EJ200 air-breathing engines, on the dry mass of the first stage and on the base drag of the first stage. OpenURNS [42] library is used for the uncertainty quantification study, using Monte-Carlo method. This allows to identify flight envelopes for both ascent and RTLS phase [43,44], and especially for the final phase in order to evaluate the safety of the return trajectories (Fig. 30).

Eventually, a sensitivity analysis is conducted in order to identify which uncertainty has the greatest impact on the overall performance

(Fig. 31), meaning on the variance of the resulting GLOW. This is achieved by computing Sobol' indices from Polynomial Chaos Expansions built on the dispersed trajectories data [45] combined with Karhunen–Loève reduction dimension [43,44]. It appears that the uncertainty on the specific impulse of the second stage engine has the most effect of the dispersion of the performance of the vehicle, following by the uncertainty on the specific impulse of the first stage. All the other uncertainties have negligible impacts and could be frozen in a first step. This study allows to identify which model uncertainty (in this case, the uncertainty about the PROMETHEUS specific impulse) has to be refined in order to reduce the uncertainty of the RLV performance.

6. Conclusion

This paper focused on the design of a reusability kits for reusable launch vehicles based on winged architectures. This consists in adding key subsystems to an expandable configuration (e.g., lifting surfaces, gears, turbojet engines) to provide the main core with reusability capability. Two different families of launch vehicles have been studied in this work: the glide-back and the fly-back configurations. In order to carry out trade-off assessments between these two types of concepts, a multidisciplinary design analysis and optimization process has been implemented by coupling different key disciplines such as propulsion, geometry and structure, aerodynamics and trajectory. Then, different analyses have been performed on the glide-back and fly-back configurations to assess the most interesting concepts in terms of airfoil for the lifting surfaces, planform configurations, turbojet number and type for the fly-back concept, etc. An important aspect to be noticed is that the performances are determined globally by optimizing jointly both the ascent and Return To Launch Site phases of the trajectory,

leading to trade-off between the phases. For the same mission specifications, glide-back and fly-back concepts have been optimized and analyzed. The fly-back concept seems to be the most promising one regarding the overall Gross-Lift-Off-Weight performance (around 70t of difference). This fly-back concept has then been further investigated by improving the disciplinary models (including advanced aerodynamics calculations, thermodynamics cycle analysis for the turbojet engines, refined structural mass model, etc.). Moreover, robustness analysis has been carried out considering different model uncertainties to determine the flight envelopes under uncertainty, to ensure the robustness of the performance and to identify the most influential mis-knowledge to be further refined in the disciplinary models. In this study, the GLOW has been chosen as the main driver for the optimization. In addition to the GLOW, other criteria such as the cost, the reliability, the manufacturing complexity should be also considered in the trade-off in a multi-criteria analysis. Furthermore, in this study, several design variables of the launch vehicle such as the main core diameter or wing geometry have been considered as fixed. In future work, it could be interesting to include them in the optimization process. Furthermore, it could be interesting to analyze the possibility to use the same fuel (CH₄) for the air-breathing engine and the rocket engine for the fly-back configuration.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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